MIL-P-19161A
21 DECEMBER 1956
SUPERSEDING
MIL-P-19161(SHIPS)
28 FEBRUARY 1956

MILITARY SPECIFICATION

PLASTIC SHEET, LAMINATED, GLASS CLOTH POLYTETRAFLUOROETHYLENE RESIN

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers glass cloth, polytetrafluoroethylene resin, general purpose (type GTE) plastic sheet material primarily for electrical insulating purposes.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

L-P-406 — Plastics, Organic; General Specifications, Test Methods.

NN-B-591 — Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

MILITARY

JAN-P-103 — Packaging and Packing for Overseas Shipment — Boxes; Wood Cleated; Solid Fiberboard.

MIL-B-10377 — Boxes: Wood-Cleated, Veneer Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

FSC 9330

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AMERICAN SOCIETY FOR TESTING MATERIALS

D-150 — Book of Standards.

(Application for copies should be addressed to American Society for Testing Materials, 1916 Race St., Philadelphia 3, Pa.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Qualification. Glass cloth, polytetrafluoroethylene resin sheets shall be a product which has been tested and has passed the qualification tests specified herein (see 6.3).

3.2 Material.

3.2.1 *Construction.* The material shall consist of a glass cloth properly impregnated and bonded with a polytetrafluoroethylene resin compound or binder, processed to conform to this specification.

3.2.2 *Property values.* The material shall conform to the property values specified herein (see tables IV, V, and VI). The values obtained for each set of specimens taken from the same sheet shall be averaged before comparison with the applicable table.

3.2.3 *Uniformity.* All sheets of any lot shall be uniform in texture, finish, and specified properties.

3.2.4 *Surface defects.* The material shall be free from blisters, wrinkles, or cracks, and reasonably free from other small defects such as scratches, dents and heat marks. The material may contain streaks due to differential coloration of various warp or filler threads of the glass fabric.

3.2.5 *Warp or twist.* The warp or twist of material as delivered shall not exceed that shown in table I (see 4.8.9). Percentage of warp is given in terms of the lateral dimensions (length and width) of the material and percentage of twist is given in terms of dimensions from one corner to the opposite corner.

TABLE I. *Warp or twist.*

Thickness	Permissible variation, on basis of 36-inch dimensions[1]
Inch	Maximum percent
1/32 to under 1/16	5.00
1/16 to under 1/8	2.50
1/8 to 1/4, inclusive	1.00

[1] These values do not apply to cut pieces but only to sheet sizes as manufactured. For any standard sheet size other than 36 inches, use the following formula:

$$\frac{D_x}{D_{36}} = \frac{L_x 2}{(36)} 2$$

Where:
D_x = Permissible deviation from straight edge in inches for the given length.
D_{36} = Permissible deviation in inches for 36 inch length.
L_x = The given length in inches.

3.3 Dimensions and tolerances.

3.3.1 *Length and width.* Unless otherwise specified by the bureau or agency concerned, the manufacturer's standard sizes between 16 and 50 inches in width and between 36 and 96 inches in length will be acceptable. The length and width of sheets may vary 1 inch over or under the manufacturer's standard size. Nonuniform standard sheet dimensions caused by cutting specimens for test required by this specification shall not be cause for rejection, unless particular dimensions are specified. When particular sheet dimensions are specified (see 6.2), the permissible variation from the specified length or width shall be as shown in table II.

TABLE II. *Permissible variations in length or width of circular sawed pieces*

Nominal thickness	Permissible variation in length or width, in inches (±)		
	6 inches and under	Over 6 to under 24 inches	24 inches and over
Inch			
Up to 1/4, inclusive	0.010	0.015	1/32

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3.3.2 *Thickness.* Sheets shall be furnished in the nominal thicknesses shown in table III, as specified (see 6.2). At least 90 percent of the area of the sheet shall be within the variations shown in table III, and at no point shall the thickness as measured vary from the nominal thickness by a value greater than 125 percent of the permissible variations.

TABLE III. *Thicknesses.*[1]

Nominal thickness		Permissible variations in inch (±)
Inch	Decimal equivalent in inch	
1/32	0.031	0.0065
3/64	.047	.0075
1/16	.0625	.0075
3/32	.094	.009
1/8	.125	.012
5/32	.156	.015
3/16	.1875	.019
7/32	.219	.021
1/4	.250	.023

[1] On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

3.4 Machinability. The material shall be such that it can be drilled, tapped, sawed, and machined in all directions[1] in accordance with the manufacturer's recommended technique without cracking, splitting, or otherwise impairing the material for general use.

[1] The nature of laminated materials necessitates special precaution when drilling and tapping parallel to laminations.

3.5 Surface finish. Unless otherwise specified by the bureau or agency concerned, the surface finish of sheets shall be semi-gloss as produced by the laminating operation.

3.6 Marking. Each full-size sheet shall be legibly marked with the manufacturer's name or trademark, and the type. The method of marking shall be satisfactory to the bureau or agency concerned.

3.7 Workmanship. Laminated materials shall be manufactured and processed in a careful and workmanlike manner, and shall be free from defects which may affect the appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests. The methods of sampling, inspection and tests shall be classified as follows:

(a) Qualification tests (see 4.2).

(b) Periodic control tests (see 4.3).

(c) Lot acceptance inspection (see 4.4).

(d) Lot acceptance tests (see 4.5).

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in table IV.

TABLE IV. *Qualification tests for property values*

Property to be tested	Test paragraph	Number of specimens for each thickness	Tests per specimen	Conditioning (see 4.7)	Unit of value	Values required for each example thickness	
						1/32 inch	1/8 inch
Dielectric breakdown parallel to laminations, step-by-step test	4.8.2	11	1	A	Minimum kv.	22.0	22.0
		24	1				
		11	1	D-48/50		10.0	10.0
		24	1				
Impact strength tested edgewise:							
Cut lengthwise	4.8.3	4	1	E-48/50	Minimum ft.-lbs. per inch	2	5.0
Cut crosswise		4	1			2	4.5

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TABLE IV. *Qualification tests for property values* (Continued)

Property to be tested	Test paragraph	Number of specimens for each thickness	Tests per specimen	Conditioning (see 4.7)	Unit of value	Values required for each sample thickness	
						1/16 inch	1/8 inch
Flexural strength, tested flatwise:							
Cut lengthwise	4.8.4	4	1	A	Minimum p.s.i.	10,000	10,000
Cut crosswise		4	1			9,000	9,000
Water absorption	4.8.5	4	1	Precondition at E-1/105 condition at D_1-24/23	Maximum percent	0.25	0.25
Dielectric constant at 1 megacycle	4.8.6	4	1	D-24/23	Maximum	[3]	3.0
Dissipation factor at 1 megacycle	4.8.6	4	1	D-24/23	Maximum	[3]	0.004
Volume resistivity	4.8.7	4	1	C-96/35/90	Minimum meg. - cm.	[3]	10,000,000
Surface resistance	4.8.7	4	1	C-96/35/90	Minimum meg.	[3]	10,000
Arc resistance	4.8.8	4	1	D-48/50	Minimum seconds	180	180

[1] Short time.
[2] Step-by-step.
[3] Not required.

4.3 **Periodic control tests.** Periodic control tests and sample thicknesses of material required for the tests shall be as shown in table V.

TABLE V. *Periodic control tests for property values.*

Property to be tested	Test paragraph	Number of specimens for each thickness	Tests per specimen	Conditioning (see 4.7)	Unit of value	Values required for each sample thickness	
						1/16 inch	1/8 inch
Dielectric breakdown parallel to laminations, step-by-step test	4.8.2	[1] 1	1	A	Minimum kv.	32.0	32.0
		[2] 4	1				
Flexural strength, tested flatwise:							
Cut lengthwise	4.8.4	4	1	A	Minimum p.s.i.	10,000	10,000
Cut crosswise		4	1			9,000	9,000
Water absorption	4.8.5	4	1	Precondition at E-1/105; condition at D_1-24/23	Maximum percent	0.25	0.25

[1] Short time.
[2] Step-by-step.

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4.3.1 *Place of tests.* Periodic control tests shall be conducted at a laboratory satisfactory to the Bureau of Ships (see 6.3) at intervals of 1 year after qualification tests have been completed. The material for test shall be taken either from production runs or from material manufactured by techniques representative of production runs. The material remaining after tests, if satisfactory, may be applied to a contract or order.

4.3.2 *Reports of tests.* Five copies of the periodic control test report shall be forwarded to the Bureau of Ships (see 6.3) via the Government inspector, who shall certify that tests were conducted under his supervision and in accordance with the test procedures specified in this specification.

4.3.3 *Approval action on tests.* Upon receipt by the Bureau of Ships (see 6.3) of the periodic control test report, action to maintain or withdraw the approval will be taken on the basis of the test results. If periodic control test results are not received by the Bureau of Ships at the expiration of the yearly period, the approval will automatically be withdrawn, and the product removed from the qualified products list.

4.4 Lot acceptance inspection. Each sheet of every lot shall be subjected to surface inspection to determine compliance with section 3, except 3.1 and 3.2.2. If any of the sheets fails to conform to any of these requirements, such sheet shall be rejected.

4.4.1 Each shipping box, with its load of plastic sheets in place, shall be inspected before the box is closed, to verify conformance to all of the requirements of section 5 of this specification which do not involve test.

4.5 Lot acceptance tests.

4.5.1 Lot acceptance tests shall be conducted on material selected from production lots offered under one or several contracts or orders.

4.5.2 A production lot shall consist of all sheets of a given machine run and of a particular thickness range as shown in table VI and offered for inspection at one time. (For instance, sheets of 1/16 and 3/32 inch thicknesses fall within one thickness range (0.062 to 0.094) and may be included in one production lot.) A machine run shall consist of all of the material pressed from a coating operation in which the basic resin and filler and treating conditions are the same.

TABLE VI. *Lot acceptance tests for property values.*

Property to be tested	Test paragraph	Number of specimens for each sample	Tests per specimen	Conditioning (see 4.7)	Unit of value	Values required for each nominal thickness range[3] (inches)			
						0.031	0.062 to 0.094	0.095 to 0.125	0.126 to 0.250
Dielectric breakdown parallel to laminations, step-by-step test	4.8.2	[1] 1 [2] 4	1 1	A	Minimum kv.	32.0	32.0	32.0	32.0
Flexural strength, tested flatwise:									
Cut lengthwise	4.8.4	4	1	A	Minimum p.s.i.	10,000	10,000	10,000	10,000
Cut crosswise		4	1			9,000	9,000	9,000	9,000

[1] Short time.
[2] Step-by-step.
[3] These ranges are for nominal thicknesses, subject to the tolerances specified in table III.

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4.5.3 *Sampling.* Sample sheets shall be selected in accordance with table VII and 4 specimens shall be cut from each sample sheet. All of the tests of table VI shall be performed on each specimen and the results on the four specimens shall be averaged and compared to determine conformance with the requirements of this specification.

TABLE VII. *Sampling for lot acceptance tests.*

Lot size (sheets)	Sample (sheets)
15 and under	1
16 to 25	2
26 and over	3

4.5.4 *Acceptance or rejection of lots.* If any of the sample sheets tested fail any of the tests shown in table VI the entire lot shall be rejected. At the option of the manufacturer, a rejected lot may be resubmitted for lot acceptance inspection and tests, but only after the manufacturer has subjected specimens from each sheet in the lot to the lot acceptance test wherein failure was observed, and has removed all nonconforming sheets.

4.5.5 *Action on test results.* Acceptance or rejection of lots will normally be determined on the basis of the lot acceptance tests.

4.5.6 *Report of test results.* Three copies of the lot acceptance test report, certified by the Government inspector, shall be forwarded to the bureau or agency concerned with a statement as to whether the lot was accepted or rejected.

4.6 Right of Government inspector. The Government inspector may at any time supervise and witness any or all of the tests specified in this specification.

4.7 Conditioning.

4.7.1 *Nomenclature.* The following letters shall be used to indicate the respective general conditioning procedures:

Condition A — As received; no special conditioning.

Condition C — Humidity conditioning.

Condition D — Immersion conditioning in distilled water.

Condition E — Temperature conditioning.

Note. Whenever a conditioning letter is followed by an inferior 1 as D_1 a prior temperature conditioning has been carried out.

4.7.2 *Designation.* Conditioning procedures shall be designated as follows:

(a) A capital letter indicating the general condition of the specimen, that is, as received, humidity, immersion, or temperature conditioning.

(b) A number indicating in hours the duration of the conditioning.

(c) A number indicating in degrees centigrade the conditioning temperature.

(d) A number indicating relative humidity whenever relative humidity is controlled.

The numbers shall be separated from each other by a slant mark, and from the capital letter by a dash.

Examples:

Condition C–96/35/90 — Humidity condition, 96 hours at 35° C. and 90 percent relative humidity.

Condition D–48/50 — Immersion condition, 48 hours in distilled water at 50° C.

Condition E–48/50 — Temperature condition, 48 hours at 50° C.

4.7.3 *Time tolerances.* Oven conditioning shall be followed by a cooling to room temperature (23° C.) in a desiccator, and immersion conditioning shall be followed by cooling to room temperature in distilled water, as specified in table VIII.

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TABLE VIII. *Conditioning time tolerances.*

Conditioning		Cooling		Comments
Condition	Time tolerance	Time	Time tolerance	
E-48/50	—0 hours +2 hours	16 hours or more	—0 hours	Cool in desiccator. Start test within ½ hour after removing specimen from desiccator.
D-48/50 D-24/23	—0 hours +½ hour	1 hour	—0 hours +2 hours	Cool by immersion in a sufficient quantity of distilled water to reduce the temperature to 23° C. within 1 hour. Remove individually as needed, and wipe surface water off with a cloth. Start test within 1 minute after removing specimen from water.
E-1/105	—0 minutes +6 minutes	2 hours or more	—0 hours	See water-absorption test (see 4.8.5).
D_1-24/23	—0 hours +2 hours	———	———	See water-absorption test (see 4.8.5).
C-96/35/90	—0 hours +2 hours	———	———	Tests after humidity conditioning shall be made on specimens in the humidity chamber (see 4.8.7). Forced air circulation shall be used in the humidity chamber.

4.7.4 *Temperature tolerances.* The conditioning temperature tolerances shall be as follows:

Nominal temperature Degrees C.	Tolerance (±) Degrees C.
23	2
35	1
50	2
105	2

4.7.5 *Humidity tolerance.* Tolerance on the nominal relative humidity of 90 percent shall be plus or minus 2 percent.

4.8 **Test Methods.**

4.8.1 *Definitions.* In conducting tests, the term "lengthwise" shall be interpreted to mean that sheet direction known to be the stronger in flexure. "Crosswise" shall then be the sheet direction known to be the weaker in flexure.

4.8.2 *Dielectric breakdown parallel to laminations. (step-by-step).*

4.8.2.1 *Specimens.* The test specimens shall be 2 by 3 inches by the thickness of the sheet. American Standard tapered pins having a taper of one-quarter inch per foot shall be used. For test specimens having a thickness up to ¼ inch, inclusive, No. 3 American Standard tapered pins[2] 3 inches long and having a diameter at the large end of seven thirty-seconds of an inch shall be used. Drill two $\frac{7}{16}$-inch diameter holes, centrally located, 1 inch apart, center to center, and perpendicular to the faces of the specimen. Using standard tapered pin reamer, ream holes to a sufficient depth to allow the pins to extend approximately 1½ inches from the small end of the hole. The electrodes shall be inserted after the conditioning of the test specimen. Spheres having a ½-inch diameter, when placed on the extremities of the tapered pins, will decrease the tendency to flashover.

4.8.2.2 *Procedures.* All tests shall be made under oil, at commercial power frequency of 60 cycles. One specimen only shall be tested

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by the short-time method for the purpose of providing a basis for the initial voltage applied in the step-by-step test. In the short-time test, starting at zero, the testing voltage shall be increased as uniformly as possible at a rate of 500 volts per second. On the remaining specimens, the step-by-step method shall be used. The initial voltage applied in the step-by-step test shall be 50 percent of the short-time breakdown voltage. The initial voltage in the step-by-step tests, determined as specified above, shall be applied for 1 minute and the voltage shall then be increased in increments as follows, holding the voltage at each step for 1 minute:

Breakdown voltage by short time method Kilovolts	Increment of increase in step-by-step method Kilovolts
12.5 or less	0.5
Over 12.5 to 25, inclusive	1.0
Over 25 to 50, inclusive	2.5
Over 50 to 500, inclusive	5.0

4.8.2.3 *Report.* The breakdown voltage shall be reported in kilovolts and the time required to break each specimen shall be reported in seconds.

4.8.3 *Impact strength (tested edgewise).*

4.8.3.1 *Specimens.* Specimens of sheets in thicknesses of ⅛ inch shall be tested in sizes 2½ inches by ½ inch by thickness without build-up. Each specimen shall be notched in one edge (side). The notch shall be centrally located with respect to the ends of the specimen. The angle of the notch shall be 45 degrees and the depth under the notch shall be 0.400 ±0.005 inch with a curvature at the bottom of the notch of 0.010 ±0.002 inch radius. Different sets of specimens shall be cut lengthwise and crosswise of the sheet.

4.8.3.2 *Procedure.* After the conditioning of the specimen, the impact strength shall be determined on a pendulum type Izod impact machine. Accurate corrections shall be made for friction and windage losses. The specimen shall be located in the machine by means of a jig, with center line of notch on the level of the top of the clamping surface. In the striking position, the striking edge of the pendulum shall be 0.866 inch from the edge of the specimen clamp. The pendulum shall be released from such a position that the linear velocity of the striking edge at the instant of impact shall be approximately 11 feet per second, which is the linear velocity corresponding to an initial elevation of the striking edge of 2 feet. The blow shall be struck on the notched side. The pointer setting shall be recorded and corresponding correction shall be obtained from a chart. This chart shall be constructed by measuring the friction and windage losses, using the proportionate amount of these losses as the correction factor.

4.8.3.3 *Calculations.* The impact strength shall be taken as the energy absorbed in breaking the specimen. It shall equal the difference between the energy remaining after breaking the specimens and the pendulum's initial energy, and shall be expressed in foot-pounds per inch of notch.

4.8.4 *Flexural strength (tested flatwise).*

4.8.4.1 *Specimens.* Specimens of sheet in thicknesses of 1/32 to ¼ inch, inclusive, shall be tested flatwise in accordance with method 1031 of Specification L–P–406, except that the dimensions and speed of test shown in table IX shall be used. Different sets of specimens shall be cut lengthwise and crosswise of the sheet.

TABLE IX. *Dimensions of specimen and speed of test.*

Nominal specimen thickness	Width of specimen	Length of specimen	Span	Speed of test per minute
Inch	Inch	Inches	Inches	Inch
1/32	1	2½	*⅝	0.023
1/16	1	3	1	.026
3/32	1	3½	1½	.040
⅛	1	4	2	.053
5/32	½	5	3	.080
¼	½	6	4	.106

* This span depth ratio is greater than 16 to 1 in order to give clearance between moving head and specimen supports.

[1] For information on tapered pins, see Kent's Mechanical Engineers Handbook, 12th Edition, Design and Production. Page 15-14. This publication may be obtained from John Wiley and Sons, Inc., 440 — 4th Ave., New York 16, N. Y.

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4.8.4.2 *Procedure.* The breadth and depth of the specimens shall be measured to the nearest 0.001 inch. The span length shall be measured to the nearest 0.01 inch. No modulus data need be taken. Care shall be taken that the specimens are centered properly in the jig prior to being tested.

4.8.4.3 *Calculations.* The flexural strength is expressed as maximum fiber stress and shall be calculated as follows:

$$\text{Maximum fiber stress} = \frac{3\,PL}{2bd^2}$$

Where:

P = breaking load in pounds.

L = the span in inches.

b = breadth of specimen in inches.

d = depth of specimen in inches.

4.8.5 *Water absorption.*

4.8.5.1 *Specimens.* The specimens shall be 3 inches by 1 inch by thickness.

4.8.5.2 *Procedure.* The specimens shall be weighed individually and then subjected to 105° C. for 1 hour, cooled in a desiccator for at least 2 hours, and reweighed (W_1). The specimens shall then be immersed in distilled water for 24 hours at 23° C. They shall be removed individually as needed, all surfaces wiped off with a damp cloth, and then reweighed immediately (W_2). The test shall begin not more than 1 minute after removal of the specimens from the water and shall be completed as rapidly as consistent with accuracy. Specimens $\frac{1}{16}$ inch or less in thickness shall be placed in a weighing bottle immediately after wiping and shall be weighed in the bottle. No correction shall be made for the water-soluble matter.

4.8.5.3 *Calculations.* The percentage of water absorption shall be calculated as follows:

$$\text{Percentage of water absorption in 24 hours} = \frac{W_2 - W_1}{W_1} \times 100$$

4.8.6 *Dielectric constant and dissipation factor (at 1 megacycle).*

4.8.6.1 *Specimens.* The specimens for dielectric constant and dissipation factor tests at 1 megacycle frequency, perpendicular to laminations, shall be 4-inch diameter disks or 4-inch squares cut from sheets $\frac{1}{8}$ inch in thickness.

4.8.6.2 *Procedure.* After the conditioning of the specimens, the electrical measurements shall be made by using a suitable bridge method, a twin-T impedance-measuring circuit, or a resonant-circuit substitution method. The method used shall have an accuracy of determination of dielectric constant of plus or minus 5 percent and an accuracy of determination of dissipation factor of plus or minus 5 percent but in no case closer than 0.0001. The specimens shall have been covered on both sides with metal foil using a thin layer of petrolatum as an adhesive, or with conductive silver paint electrodes. The dissipation factor shall be directly computed from the instrument readings and shall be considered nominally equivalent to the power factor.

Note: For a complete discussion of theory, apparatus, electrodes, and calculation, refer to method D150 of ASTM Book of Standards.

4.8.7 *Volume resistivity and surface resistance.*

4.8.7.1 *Specimens.* The specimens for volume-resistivity and surface-resistance tests shall be 4 by 4 inches by thickness.

4.8.7.2 *Procedure.*

4.8.7.2.1 *Volume resistivity.* The test electrodes shall be made of conductive silver paint sprayed or brushed on the surfaces of the specimen. The upper electrode shall be in the shape of a circle 2 inches in diameter. A guard ring $\frac{1}{4}$ inch wide shall be located concentrically with respect to the upper electrode and shall be spaced $\frac{1}{8}$ inch from it. The bottom electrode shall be in the shape of

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a circle 3 inches in diameter. The centers of the upper and lower electrodes shall be centered as accurately as possible so that they are on the same axis. The potentials applied shall be as shown on figure 1. The volume resistance shall be measured by means of a 500-volt megohm bridge exactly 1 minute after the current is applied. Tests following the humidity conditioning shall be made on specimens in the humidity chamber.

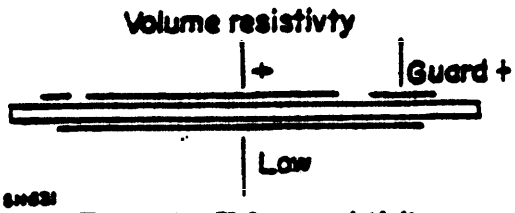


FIGURE 1. *Volume resistivity*

4.8.7.2.2 *Surface resistance.* Surface resistance measurements shall be made on the same specimens as those used for volume resistivity. The test procedure shall be similar to that specified for volume resistivity, except that the potentials of the guard and bottom electrodes shall be interchanged as shown on figure 2: The surface resistance shall be measured by means of a 500-volt megohm bridge exactly 1 minute after the current is applied. Tests following the humidity conditioning shall be made on specimens in the humidity chamber.

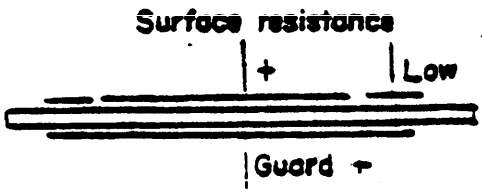


FIGURE 2. *Surface resistivity*

4.8.7.3 *Calculations.* The volume resistivity shall be calculated as follows:

$$r = \frac{RA}{L}$$

Where:

r — volume resistivity in megohm-centimeters.
R — measured volume resistance in megohms.
A — area of guarded electrode in square centimeters.
L — average thickness of specimen in centimeters.

Surface resistance shall be reported in megohms.

4.8.8 *Arc resistance.* The arc resistance test shall conform to method 4011 of Specification L-P-406.

4.8.9 *Warp and twist.* The warp and twist test shall conform to method 6051 of Specification L-P-406 (see 3.2.5).

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level A.* The laminated plastic sheets shall be wrapped individually or interleaved to protect them from abrasion, using not less than 25-pound basis weight kraft paper.

5.1.2 *Level C.* The laminated plastic sheets shall be packaged in accordance with the manufacturer's practice.

5.2 Packing.

5.2.1 *Level A.* The sheets, packaged in accordance with level A or C as specified (see 6.2), shall be packed in cleated plywood (overseas type), cleated fiberboard, or nailed wood (overseas type) boxes conforming to Specification PPP-B-601, JAN-P-108 or PPP-B-621, respectively. Corrugated fiberboard pads and liners shall be used to protect the exposed surfaces and edges of the sheets. Shipping containers shall have case liners conforming to Specification MIL-L-10547 and appendix thereto. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds unless the weight of a single sheet exceeds 200 pounds, in which case only one sheet shall be packed in a container. Fiberboard boxes shall not exceed the weight

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limitations of the applicable box specification.

5.2.2 *Level B.* The sheets, packaged in accordance with level A or C as specified (see 6.2), shall be packed in wood cleated fiberboard, cleated plywood (domestic type), wood cleated paper overlaid (domestic type), or nail wood (domestic type) conforming to Specification NN-B-591, PPP-B-601, MIL-B-10377, or PPP-B-621, respectively. Corrugated fiberboard pads and liners shall be used to protect the exposed surfaces and edges of the sheets. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds unless the weight of a single sheet exceeds 200 pounds, in which case only one sheet shall be packed in a container. Fiberboard boxes shall not exceed the weight limitations and shall conform to the special requirements of the applicable box specification.

5.2.3 *Level C.* The sheets, packaged in accordance with level A or C as specified (see 6.2), shall be packed in containers in a manner to insure safe delivery and acceptance at destination. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.3 **Marking.** In addition to any special marking specified in the contract or order, each shipping container shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 **Intended use.** Plastic material covered by this specification is intended for use in electronic applications requiring very low dielectric losses and very high volume resistivity and surface resistance.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Sheet size and thickness required (see 3.3).
(c) Levels of packaging and packing and applicable marking (see section 5).

6.3 **Qualification.** With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids been tested and approved for inclusion in Qualified Products List QPL-19161, whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C.

6.4 **Fabricated parts or equipment.** Requirements applicable to fabricated parts or equipment incorporating glass-cloth, polytetrafluoroethylene resin, laminated material should be specified in separate specifications or as part of contracts or orders.

6.5 The packaging, packing and marking specified herein apply only to direct shipments to the Government and are not intended to apply to contracts or orders between the manufacturer and the prime contractor.

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Air Force

Other interest:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM, APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-6279

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